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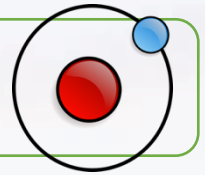
Lecture 15
Signal Emulation for Astrophysics and Cosmology

Lecturer: **Dr Harry Bevins** (htjb2)

An SKA observation

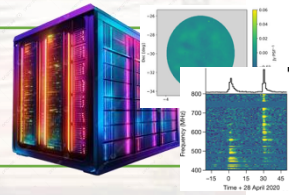
L1:
Physics



L 2,4,5:
Instrument

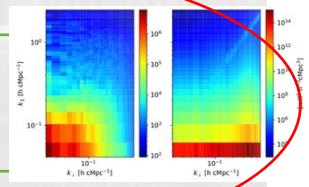
L6:
Calibration

L7-10:
Science data



L11-12:
Data handling

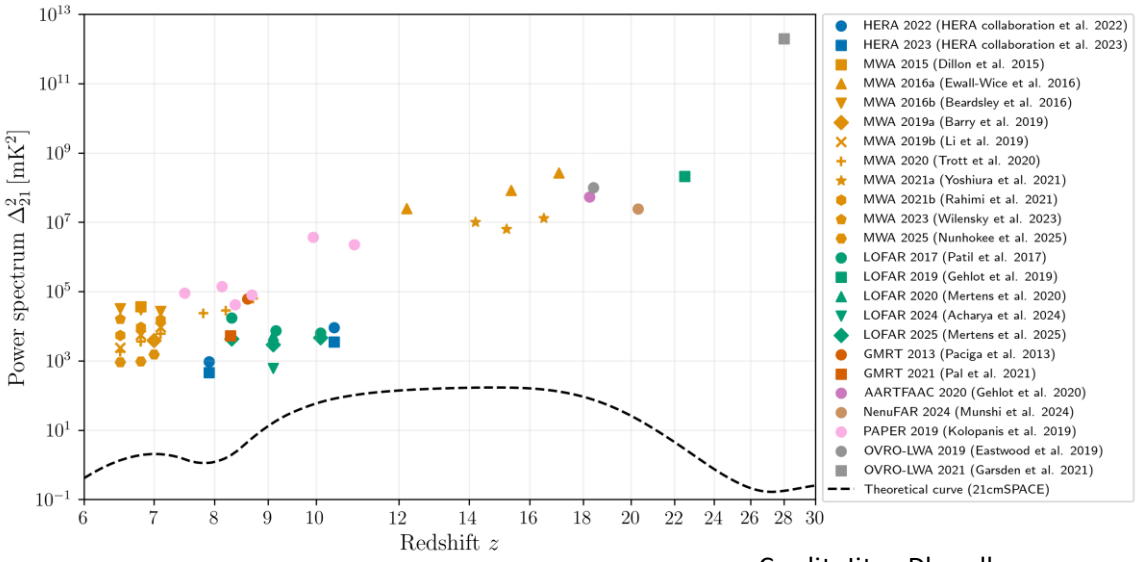
L13-16:
Science



Lectures 13-16: Science extraction

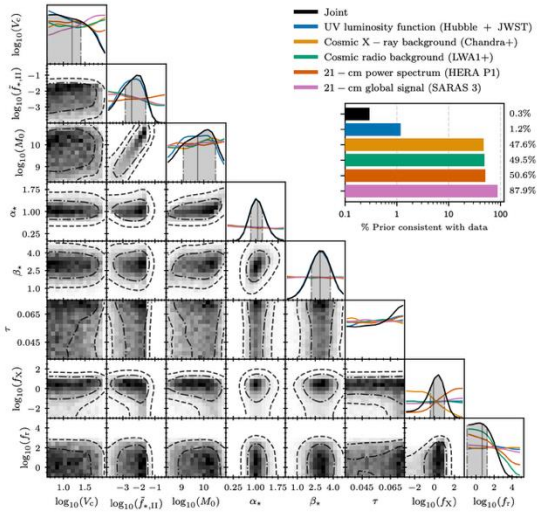
Example:

The Epoch of Re-ionization and the 21-cm line



Credit: Jiten Dhandha

From **science data products**



Dhandha et al. 2025

To **science**

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Overview

- Why we need signal emulators?
- Gaussian Processes
- Neural Networks

Slides available at https://github.com/htjb/Talks (and Moodle!)

Example codes on github too!

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Why we need signal emulators?

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Recap:Bayes theorem

Posterior: P(θ)

Evidence: Z

Likelihood: $L(\theta)$

Prior: $\pi(\theta)$

$$\underbrace{P(\theta|D)P(D)}_{\text{Outputs}} = \underbrace{P(D|\theta)P(\theta)}_{\text{Inputs}}$$

For Nested Sampling and other MCMC algorithms

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The likelihood function

- The likelihood function gives the probability of the *data given the model and parameters* $L(\theta) = P(D|\theta, M)$

- Defined as an input to Nested Sampling and other MCMC algorithms

- A lot of research goes into the form of the likelihood which varies for different analysis problems (e.g. Scheutwinkel et al. [2204.04491], Lovick et al. [2312.02075])

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An example likelihood

$$\log_e L(\theta) = \sum_i -\frac{1}{2}\log_e |2\pi\Sigma| - \frac{1}{2}\left(D_i - M_i(\theta)\right)^T \Sigma^{-1}(D_i - M_i(\theta))$$

- Work in log space to make things more computationally stable
- The functional form of the likelihood defines the distribution of the random component of your data
- Here we are assuming the noise (or random component) is Gaussian and allowing for a non-diagonal covariance

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The issue

$$\log_e L(\theta) = \sum_i -\frac{1}{2}\log_e|2\pi\Sigma| - \frac{1}{2}\left(D_i - M_i(\theta)\right)^T \Sigma^{-1}(D_i - M_i(\theta))$$

- In an MCMC run the likelihood function is evaluated for $\sim 10^5 - 10^6$ different θs
- This means we potentially need to evaluate the model $M(\theta)$ millions of times
- But the models are often computationally expensive taking minutes to hours to days per realization

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The issue

$$\log_e L(\theta) = \sum_i -\frac{1}{2}\log_e|2\pi\Sigma| - \frac{1}{2}\left(D_i - M_i(\theta)\right)^T \Sigma^{-1}(D_i - M_i(\theta))$$

- How do we get around this?
- Look towards emulators which can typically approximate $M(\theta)$ in milliseconds
- Go from millions of hours to hours or even minutes per MCMC chain

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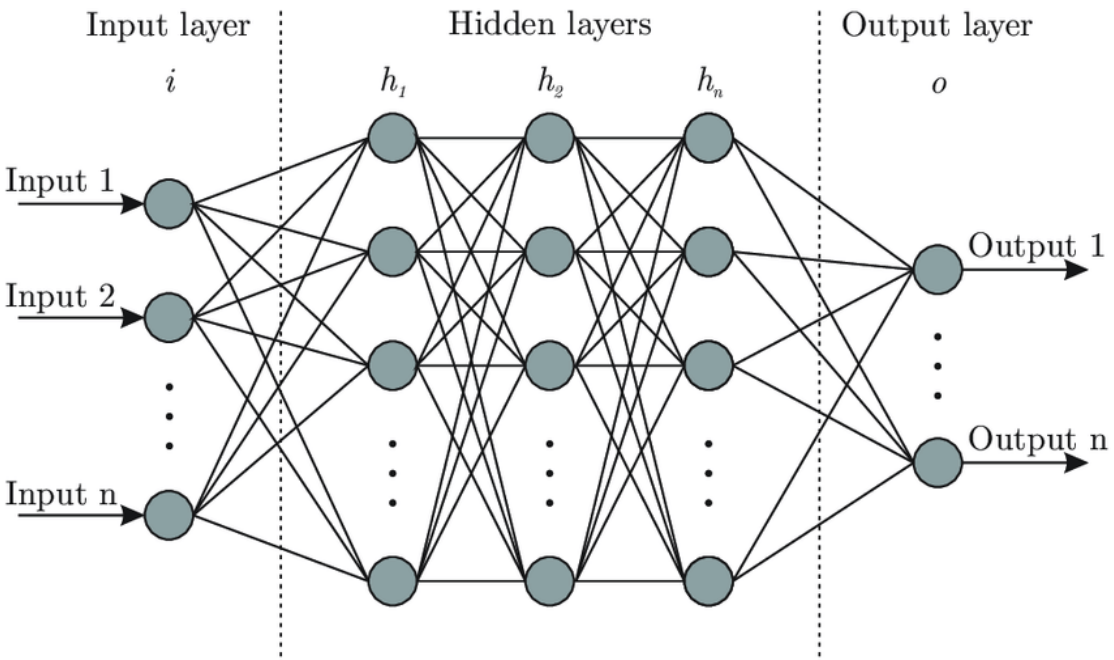
What actually is an emulator?

- Some approximation of a complex astrophysical model (think N-body, hydrodynamical or semi-numerical simulations)

- Built with machine learning tools like ANNs, CNNs and GPs



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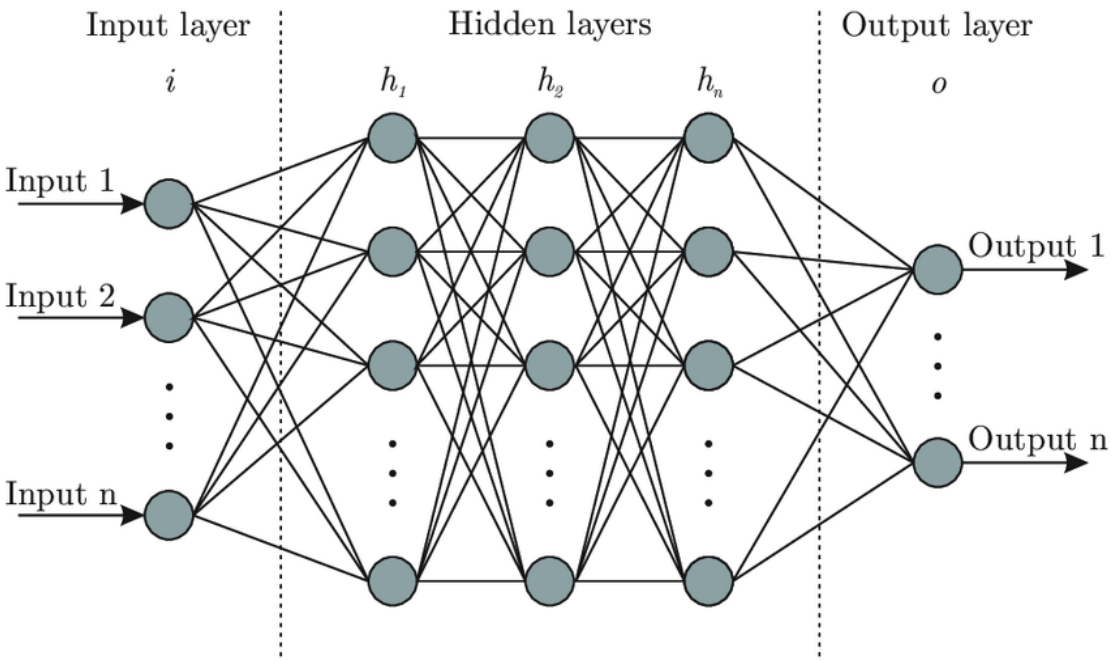
Emulators as an approximation

- In $g(\theta) \approx f_{\phi}(\theta)$, ϕ are the parameters of our emulator
- ϕ has to be optimized so that the approximation is accurate as possible
- Can say $g(\theta) = f_{\phi}(\theta) + \epsilon_{\phi}(\theta)$ where we are attempting to minimize the error $\epsilon_{\phi}(\theta)$



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21-cm Cosmology

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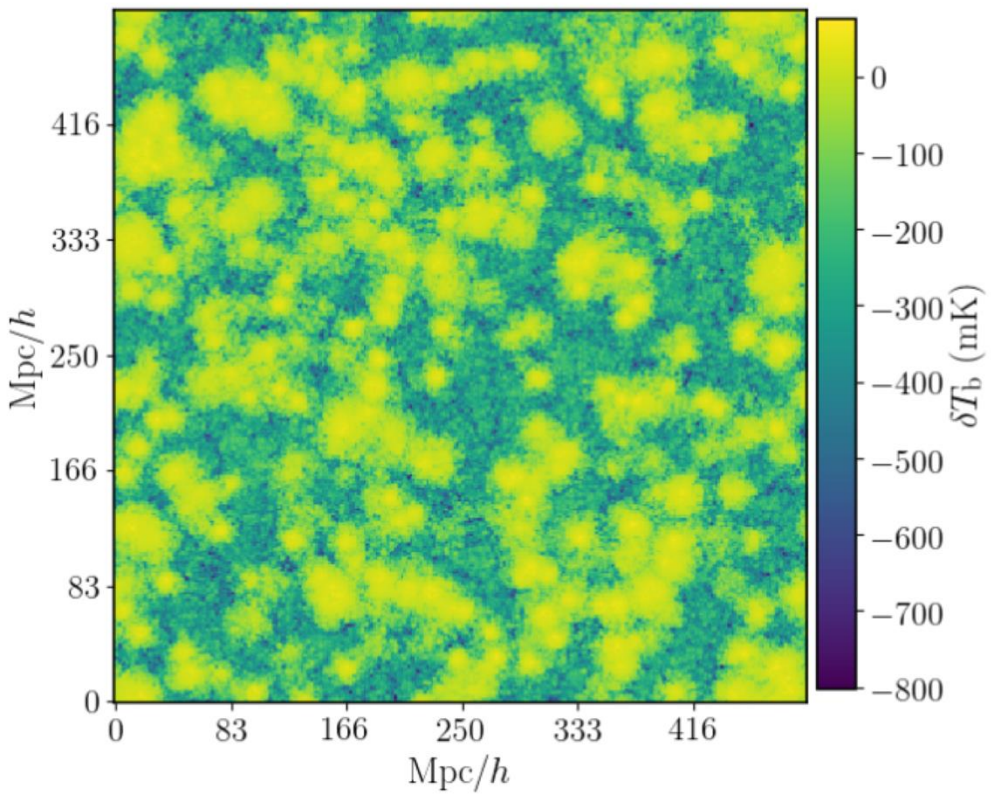
21-cm Cosmology

- Tracing the evolution of the universe through observations of the 21-cm signal from neutral hydrogen

- A detection of the signal will help us understand
  - When the first stars formed ad how bright they were
  - The nature of dark matter
  - The abundance and brightness of X-ray emitting objects
  - When the universe transformed from neutral to ionized



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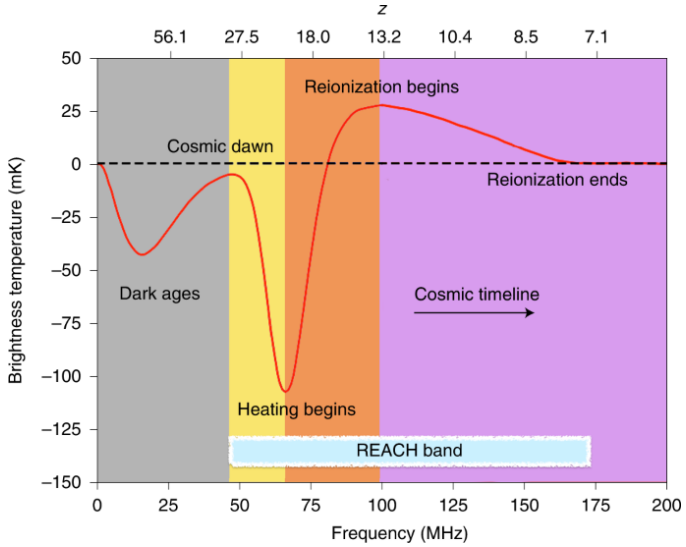
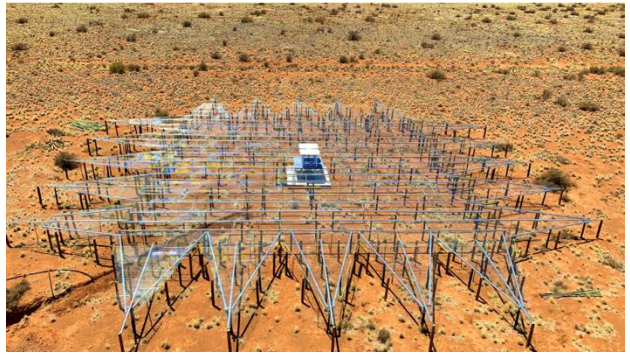
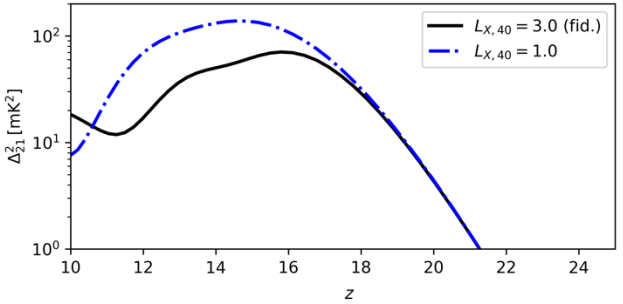
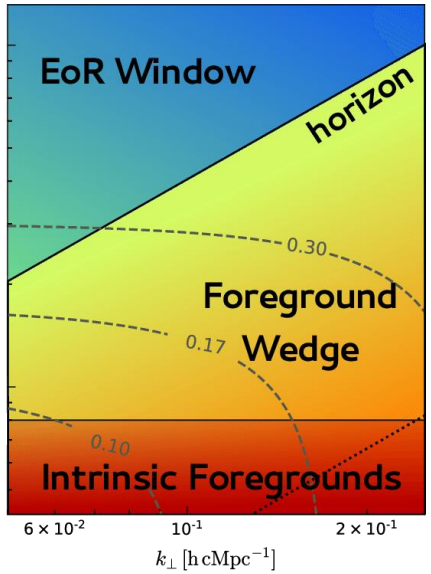
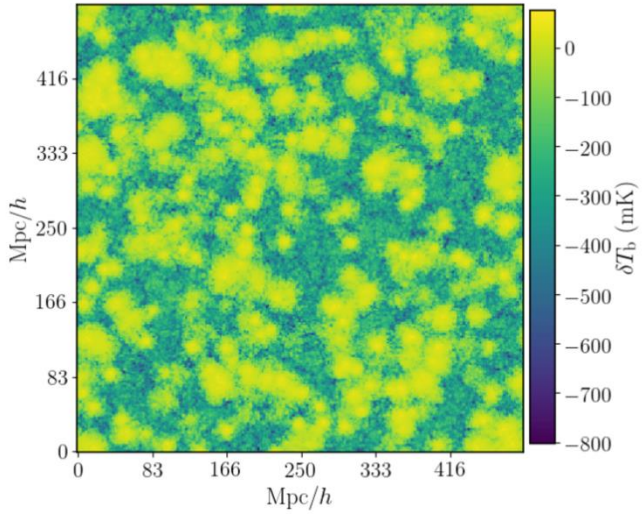
21-cm Cosmology











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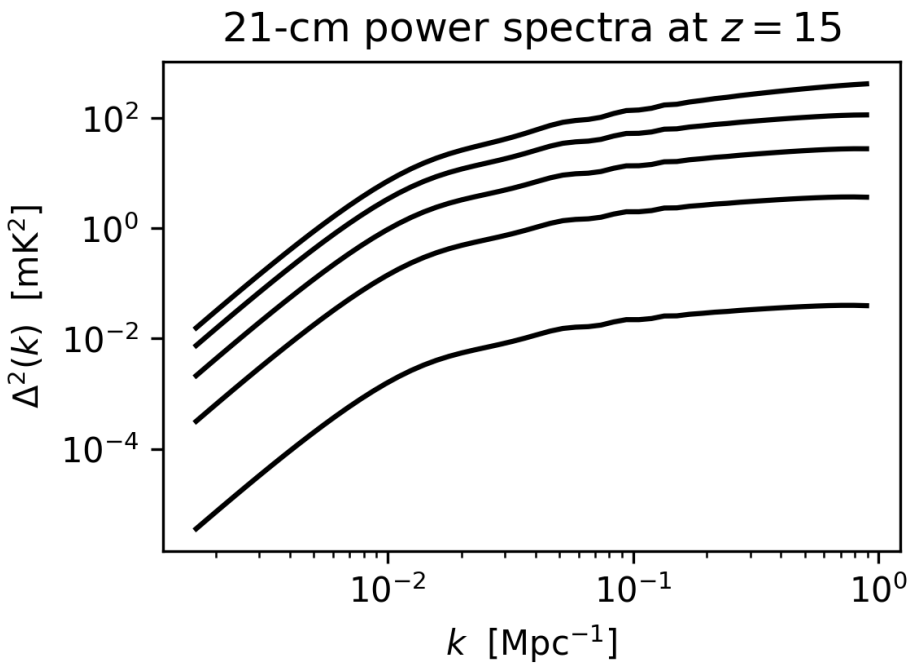
21-cm Cosmology

- Complex dependence between $\Delta^2(k)$ and the parameters of our models θ
- We use simulations to model and study how the signal varies over space and time for given θ
- Here we are using Zeus21 which is a fast approximate model of the 21-cm signal over time
- Zeus21 takes < second to run per realization however more accurate semi-numerical codes takes $\approx$ 3hrs



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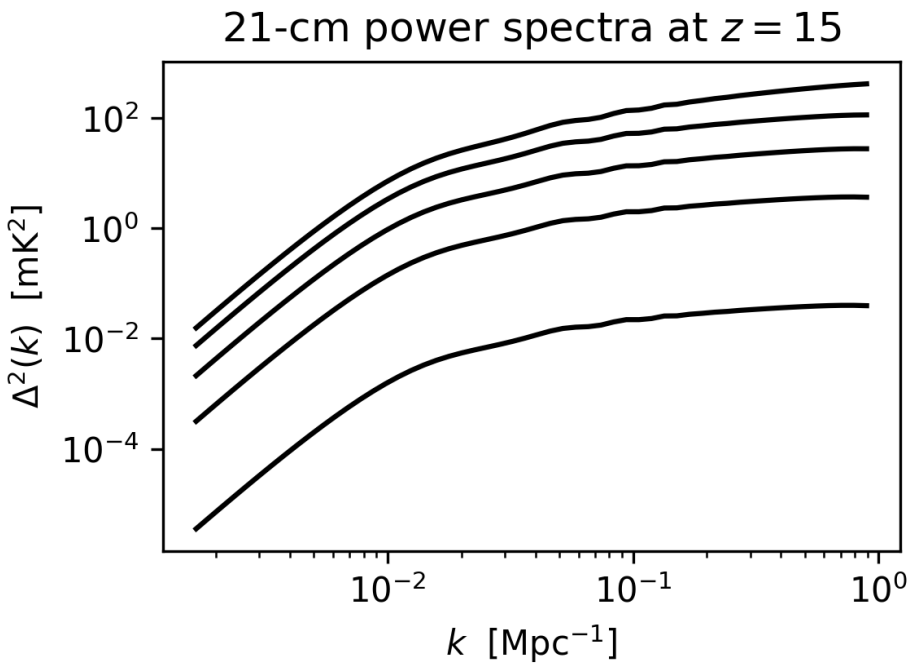
21-cm Cosmology

- The zeus21 model has many different parameters
- There are 10,000 example signals to use to build our emulators
- For the examples we are fixing a lot of them and just looking at the effects of
  - $L_{X,40}$ - X-ray luminosity
  - ϵ – star formation efficiency
  - $f_{esc,10}$ - escape fraction of ionized photons
  - h_0 - Hubble's constant



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Gaussian Process

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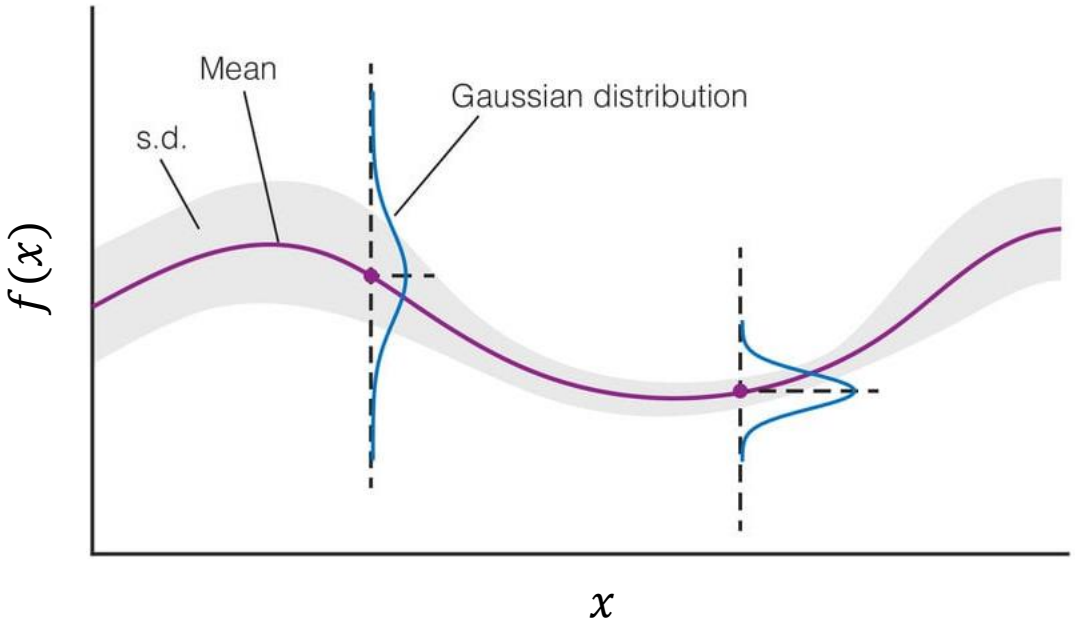
What is a Gaussian Process?

- The idea is to parameterize the distribution of all functions with a Gaussian

$$\hat{f}(x) \sim N\big(\mu(x), C(x, x')\big)$$

- $\mu(x)$ is the mean function
- $C(x, x')$ is the covariance
- Typically see this in regression problems as a non-parameteric or model independent fitting technique



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How do we use it to emulate a model?

- We have a set of simulations $f(x, \theta)$

- We decompose our data such that we have

$$f(x, \theta) = \sum_i^N w_i(\theta) g(x)$$

where w_i are weights and $g(x)$ is independent of the parameters

- We can then define N GPs

$$w_i(\theta) \sim N\big(\mu_i(\theta), C_i(\theta, \theta')\big)$$

where μ_i and C_i can be parameterized by ϕ and ϕ is optimized with data examples

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Dimensionality Reduction

- This idea

$$f(x,\theta) = \sum_{i}^{N} w_i(\theta) g(x)$$

can be very powerful

- If we make a good choice for $g(x)$ then the dimensionality of our problem reduces from N_x to N_w

- These techniques can lead to information loss, but it can be managed

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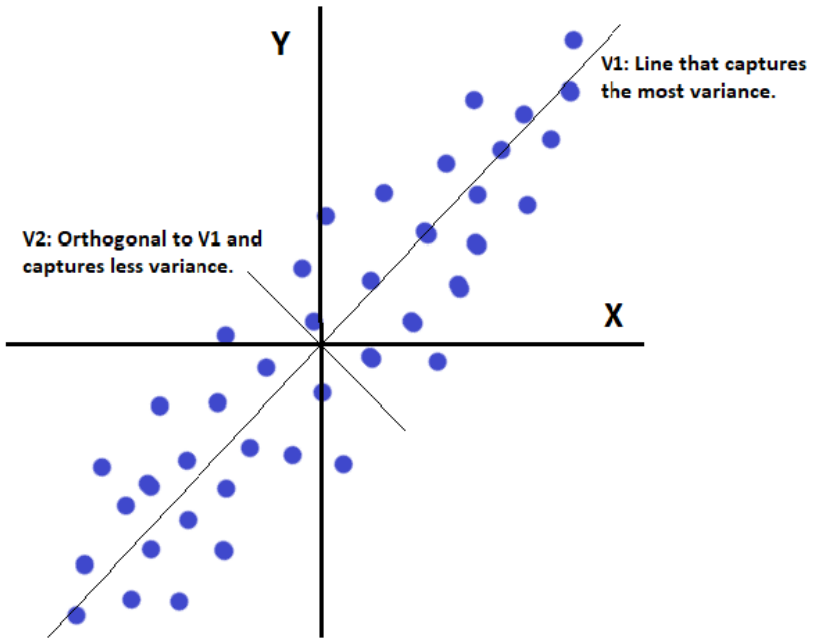
Principle Component Analysis

- Decompose training data into eigenvectors (e.g. $g(x)$) and order based on eigenvalues
- Eigenvectors with largest eigenvalues describe the directions in the training data with the biggest variance
- Discard eigenvectors that correspond to low variance directions
- Reconstruct signals with weighted sum (e.g. $w_i(\theta)$) of most important eigenvectors where weights vary for each signal



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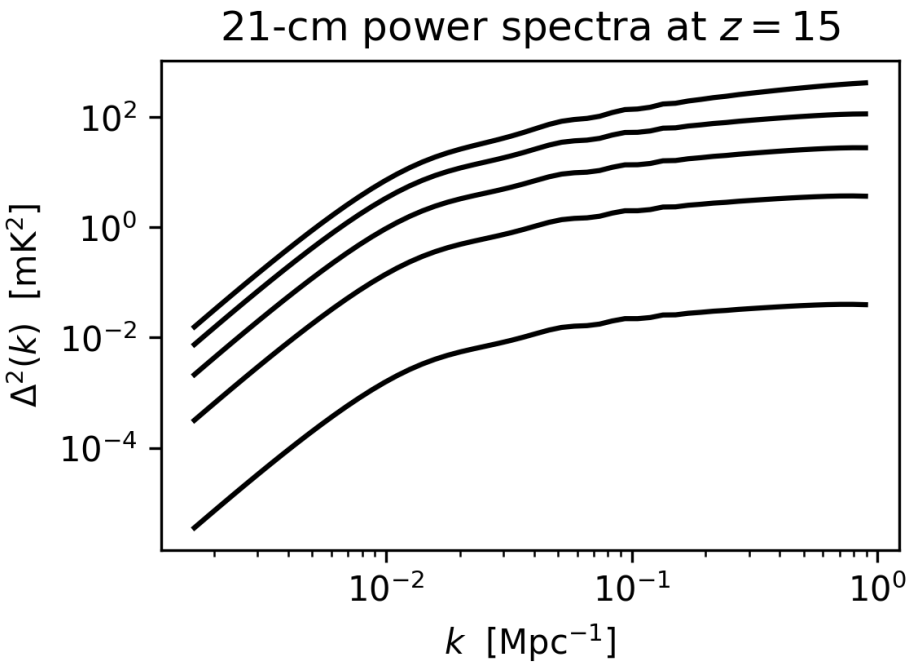
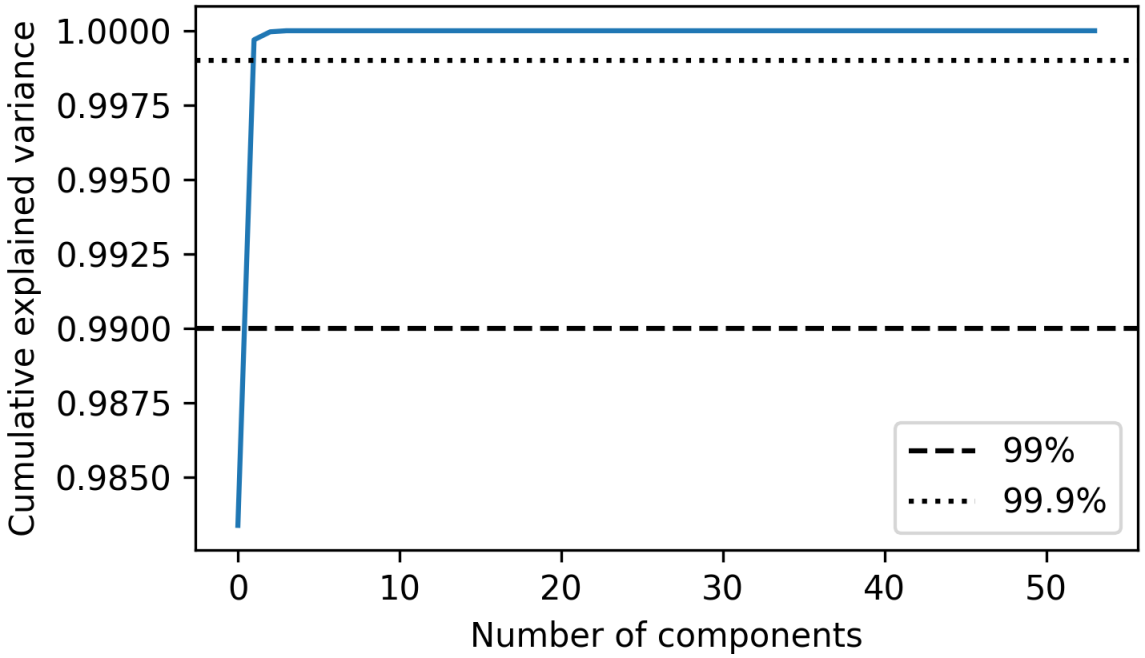
Principle Component Analysis

Cumulative explained variance for m eigenvectors out of N → $$CEV = \frac{\sum_{k=0}^{m} \lambda_k}{\sum_{k=0}^{N} \lambda_k}$$





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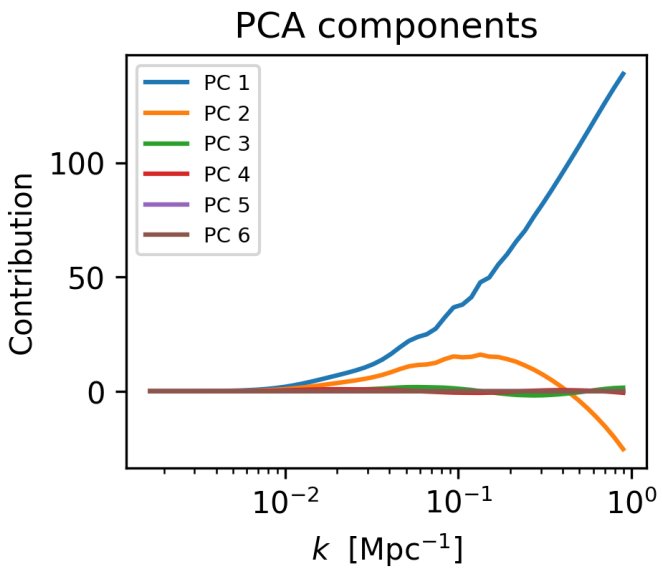
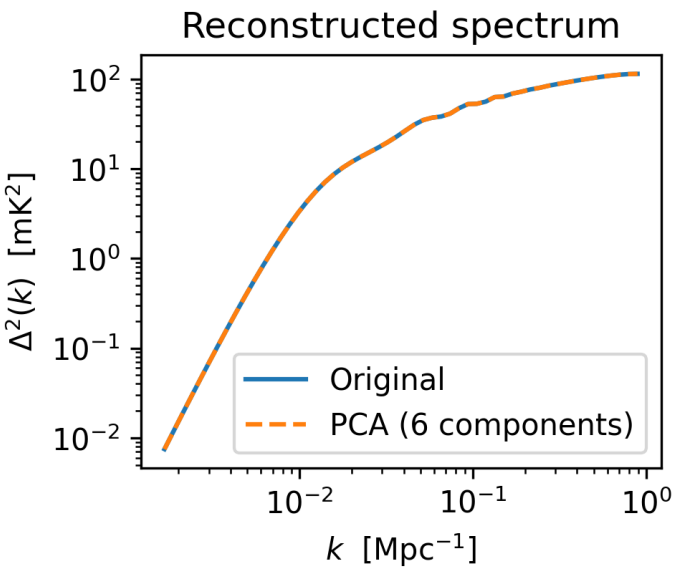
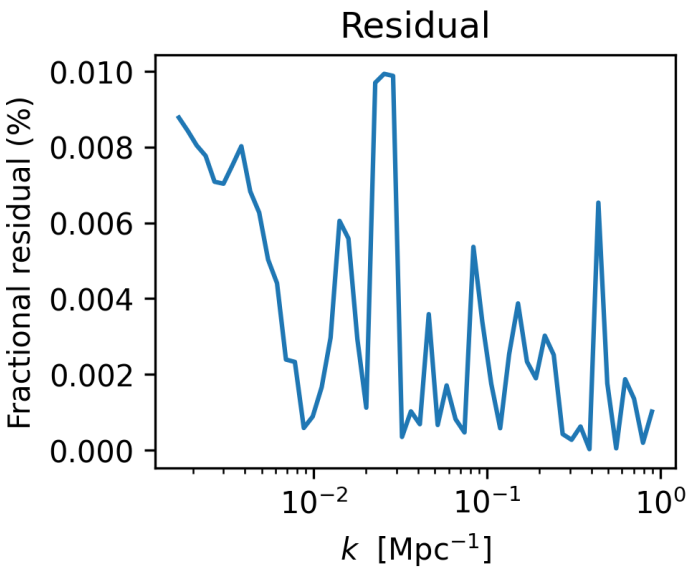
Principle Component Analysis







$$g(x) = g(k)$$

$$f(x,\theta) = \sum_i^N w_i(\theta)\, g(k)$$

$$FR = 100 \times \frac{|f(x,\theta) - \Delta^2(k)|}{|\Delta^2(k)|}$$

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GP Regression with scikit learn

- We perform a PCA decomposition on our training data and find we need 3 components to explain our data set
- We therefore want 3 GPs to predict the weights as a function of the model parameters
- Define our kernel as

$$k(\theta, \theta') = \sigma^2 \exp\left(-\frac{1}{2}\sum_{d=1}^{4}\frac{(\theta_d - \theta'_d)^2}{l_d^2}\right)$$

- Goal is to maximise the likelihood of the training examples under this model

```python
kernel = ConstantKernel(1.0) * RBF(length_scale=[1.0]*4)

gps = []
for i in range(n_components):
    gp = GPR(
        kernel=kernel,
        alpha=1e-6,
        n_restarts_optimizer=5,
        random_state=i,
    )
    gp.fit(train_X, train_pca[:, i])
    gps.append(gp)
    print(f"GP {i+1} trained | log-marginal-likelihood =" +
          f"{gp.log_marginal_likelihood_value_:.2f}")
```

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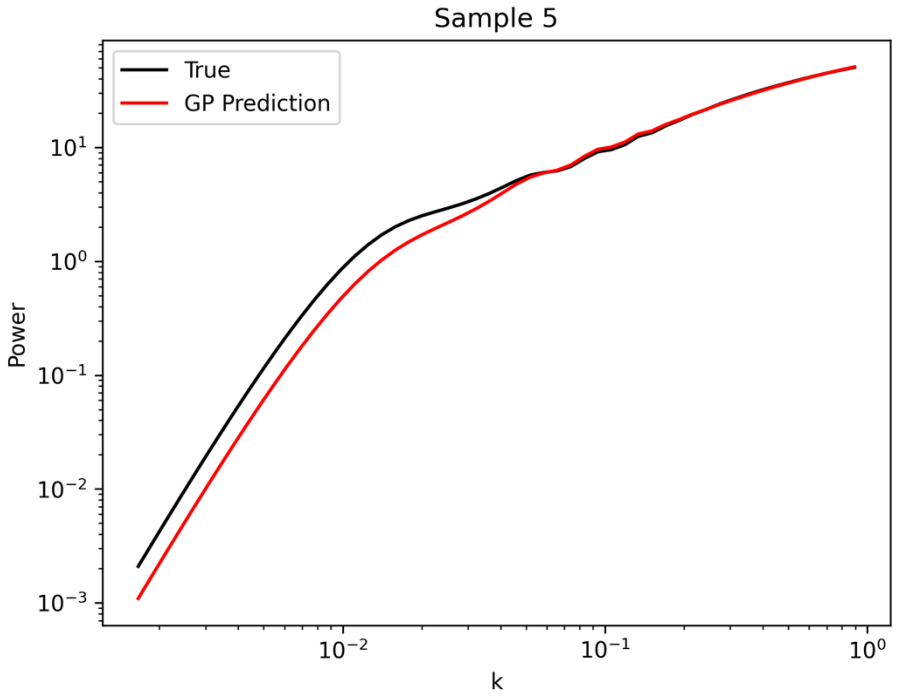
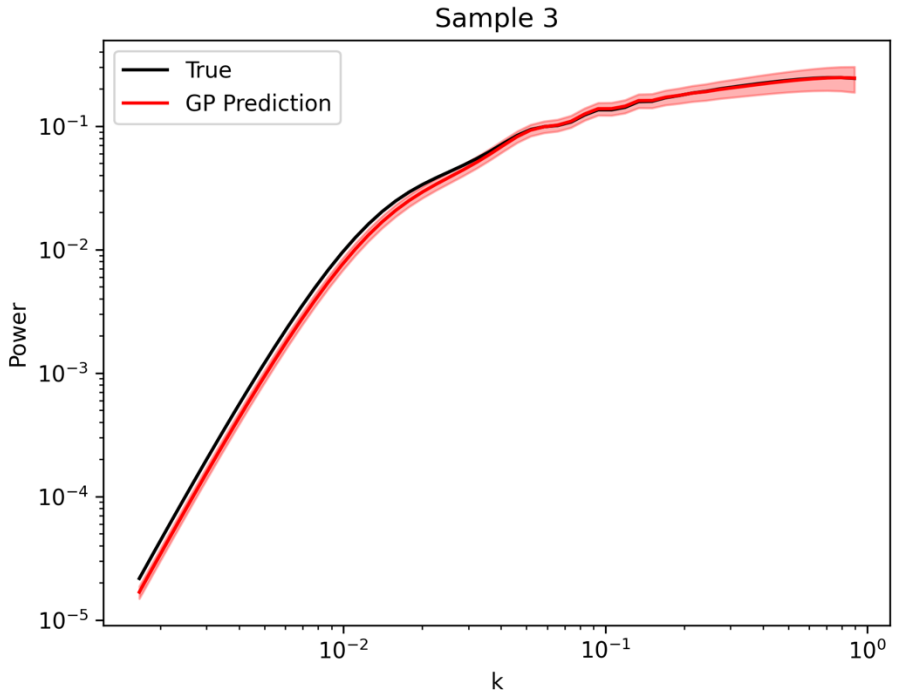
Assessing performance

$$\theta \rightarrow \text{Trained GPs} \rightarrow w_i(\theta) \text{ for } i = 0 \ldots 3 \rightarrow f(\theta, x) = \sum_i^N w_i(\theta) g(x)$$





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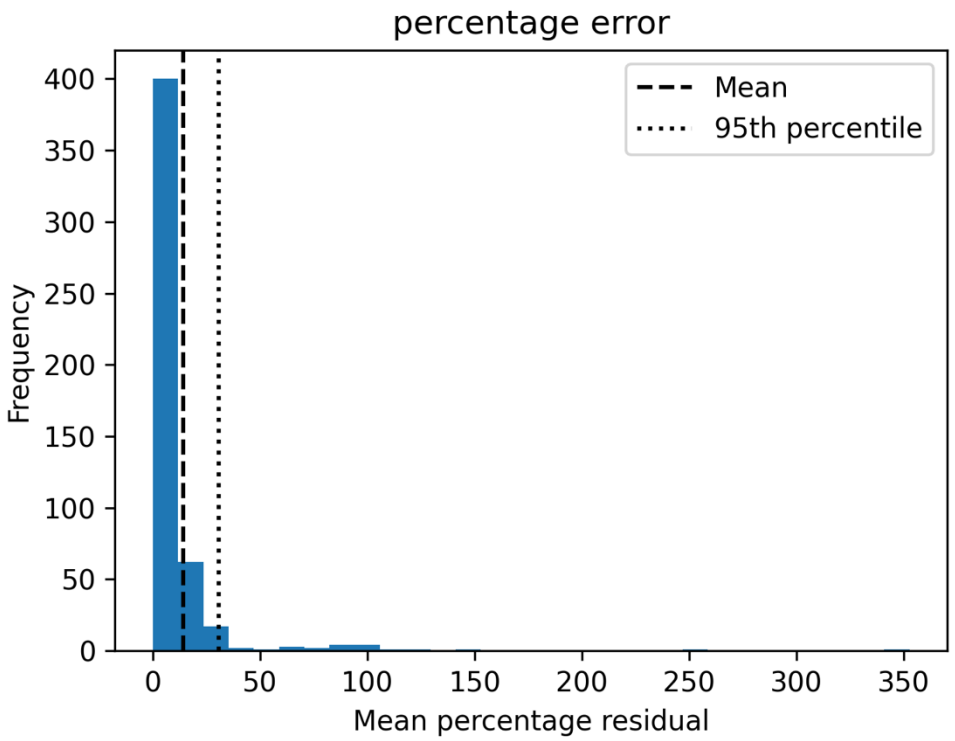
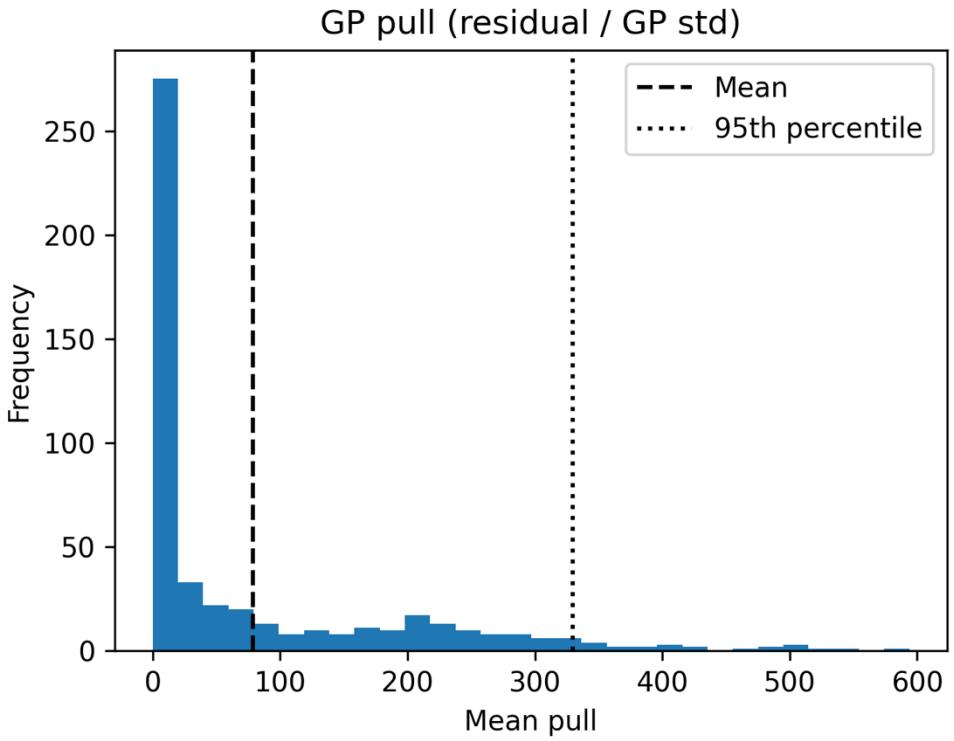
Assessing performance

$$\epsilon_i = 100\,\frac{1}{N_k}\sum_k \frac{|\Delta^2_{true}(k) - \Delta^2_{pred}(k)|}{|\Delta^2_{true}(k)|}$$

$$\epsilon_i = \frac{1}{N_k}\sum_k \frac{|\Delta^2_{true}(k) - \Delta^2_{pred}(k)|}{|\sigma_\Delta(k)|}$$





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Neural Networks

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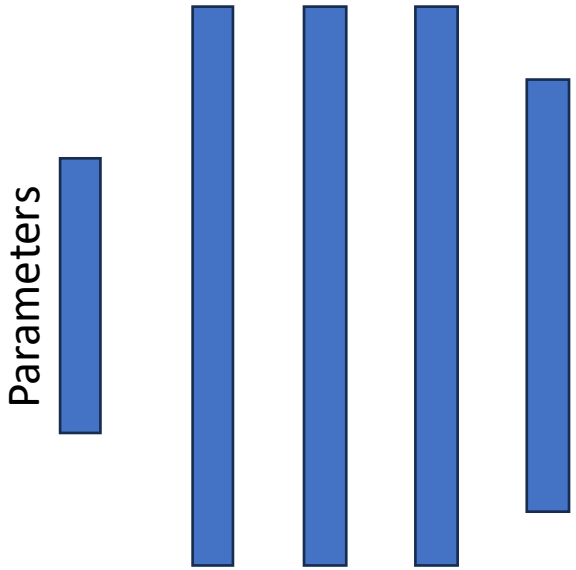
Principle Component Analysis



Signal Specific PCA coefficients $w_i(\theta)$ $\times$ Important eigenvectors $g(x)$ $\longrightarrow$ Sum Terms

$$f(x,\theta) = \sum_i^N w_i(\theta) g(x)$$

NNs are a bit more flexible than GPs so we have to think more carefully about some of our design choices

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Loss Function

- As with the GPR, we provide the network with training data $\{\Delta^2, \theta\}$ but there is a bit more freedom in our objective function
- Recall we are trying to find the best set of parameters ϕ to minimise the error in our approximation ϵ_ϕ
- Two common choices are the MAE and MSE

$$L = \frac{1}{N}\sum_i \left|\epsilon_\phi(\theta)\right| \qquad L = \frac{1}{N}\sum_i \left(\epsilon_\phi\right)^2$$

- NNs are typically optimized with some form of Stochastic Gradient descent

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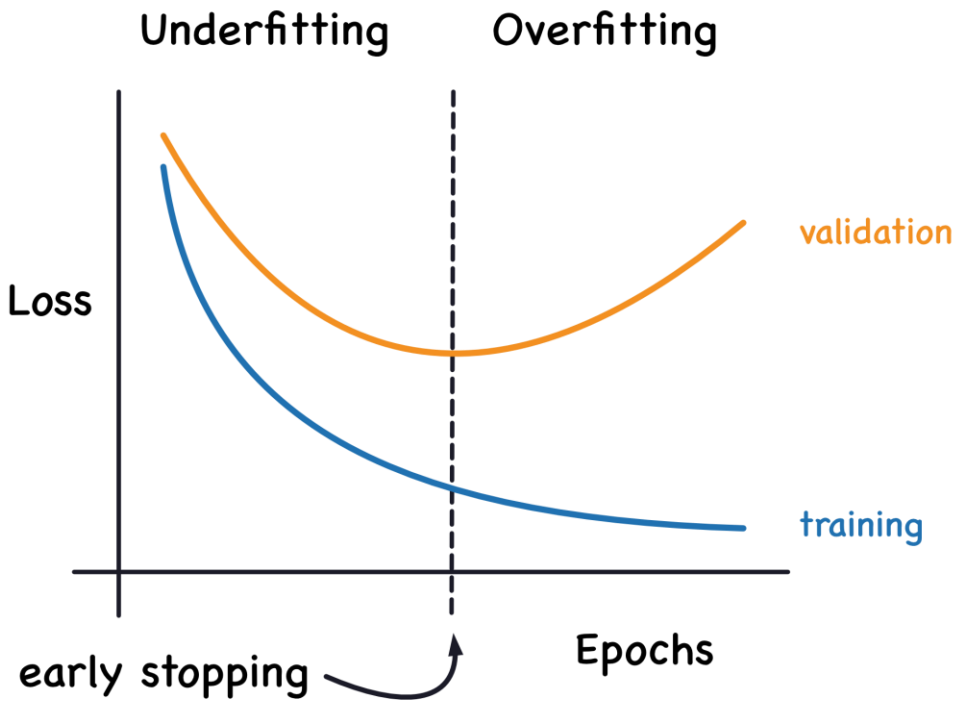
Training, test and validation data

- We reserve a proportion of our data for validating the performance of the emulator during training (early stopping)



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Building the neural network

Many frameworks for building NNs in Python including Tensorflow, JAX and PyTorch…

```
class Emulator(nn.Module):
    def __init__(self, input_dim, output_dim, hidden_dim=16, num_layers=3):
        super(Emulator, self).__init__()
        input = nn.Linear(input_dim, hidden_dim)
        output = nn.Linear(hidden_dim, output_dim)

        self.layers = nn.ModuleList()
        self.layers.append(input)
        for i in range(num_layers):
            self.layers.append(nn.Linear(hidden_dim, hidden_dim))
        self.layers.append(output)

    def forward(self, x):
        x = torch.relu(self.layers[0](x))
        for layer in self.layers[1:-1]:
            x = torch.relu(layer(x))
        x = self.layers[-1](x)
        return x
```

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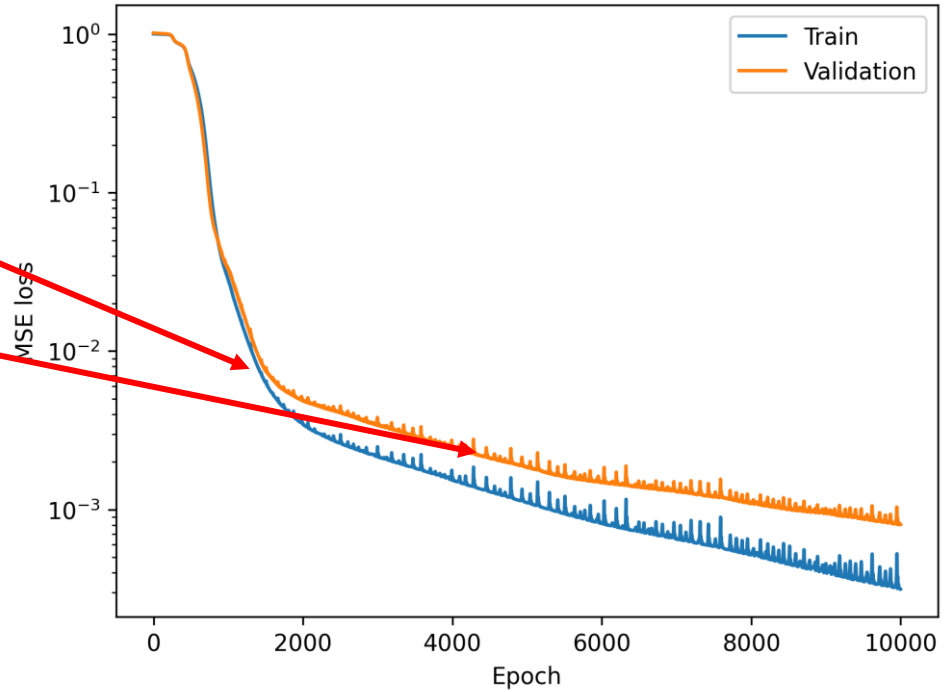
Training

```python
n_epochs = 10000
patience = 100
counter  = 0
best_val_loss    = float('inf')
best_model_state = model.state_dict()
train_losses, val_losses = [], []

pbar = tqdm(range(n_epochs))
for epoch in pbar:
    model.train()
    optimizer.zero_grad()
    loss = criterion(model(train_X_tensor), train_pca_tensor)
    loss.backward()
    optimizer.step()

    model.eval()
    with torch.no_grad():
        val_loss = criterion(model(val_X_tensor), val_pca_tensor)

    train_losses.append(loss.item())
    val_losses.append(val_loss.item())

    if val_loss < best_val_loss:
        best_val_loss    = val_loss
        best_model_state = model.state_dict()
        counter = 0
    else:
        counter += 1
        if counter >= patience:
            print(f'Early stopping at epoch {epoch+1}')
            break

    pbar.set_description(
        f'Loss: {loss.item():.4f}  Val: {val_loss.item():.4f}'
    )

model.load_state_dict(best_model_state)
```

Training Step

Validation Step

Early Stopping



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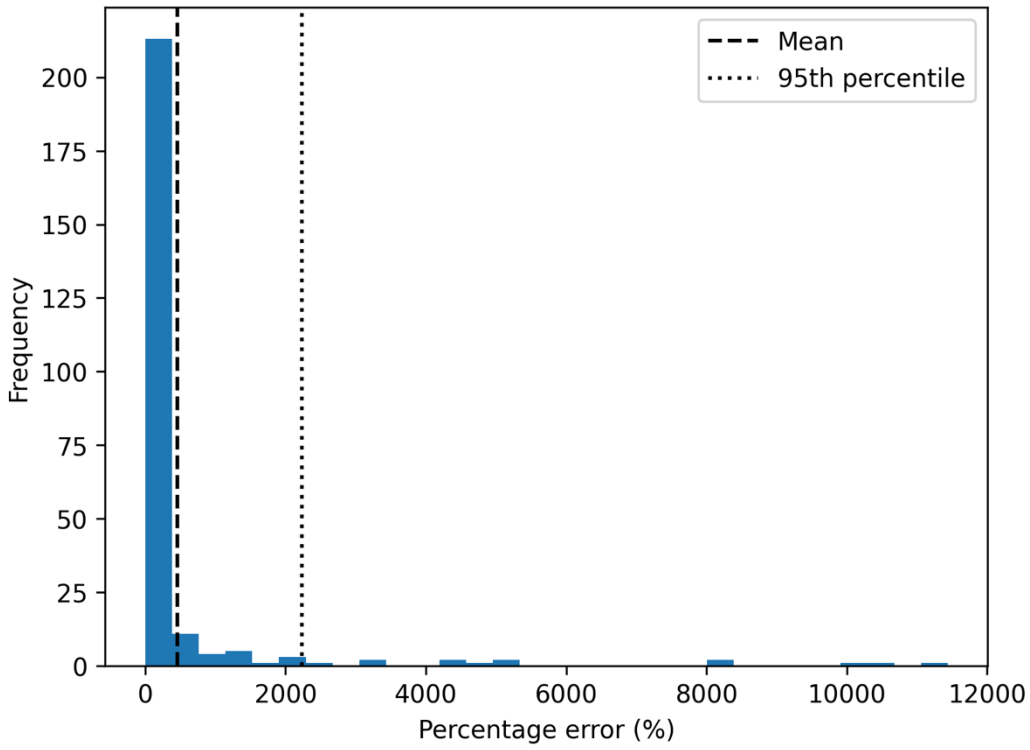
Assessing the accuracy

$$\epsilon_i = 100 \frac{1}{N_k} \sum_k \frac{|\Delta^2_{true}(k) - \Delta^2_{pred}(k)|}{|\Delta^2_{true}(k)|}$$

- We assess on the whole test data set and take the average and 95th percentile values

- In this case $\bar{\epsilon} \approx 456.693$ % and $\epsilon_{95} \approx 2230.327$ %



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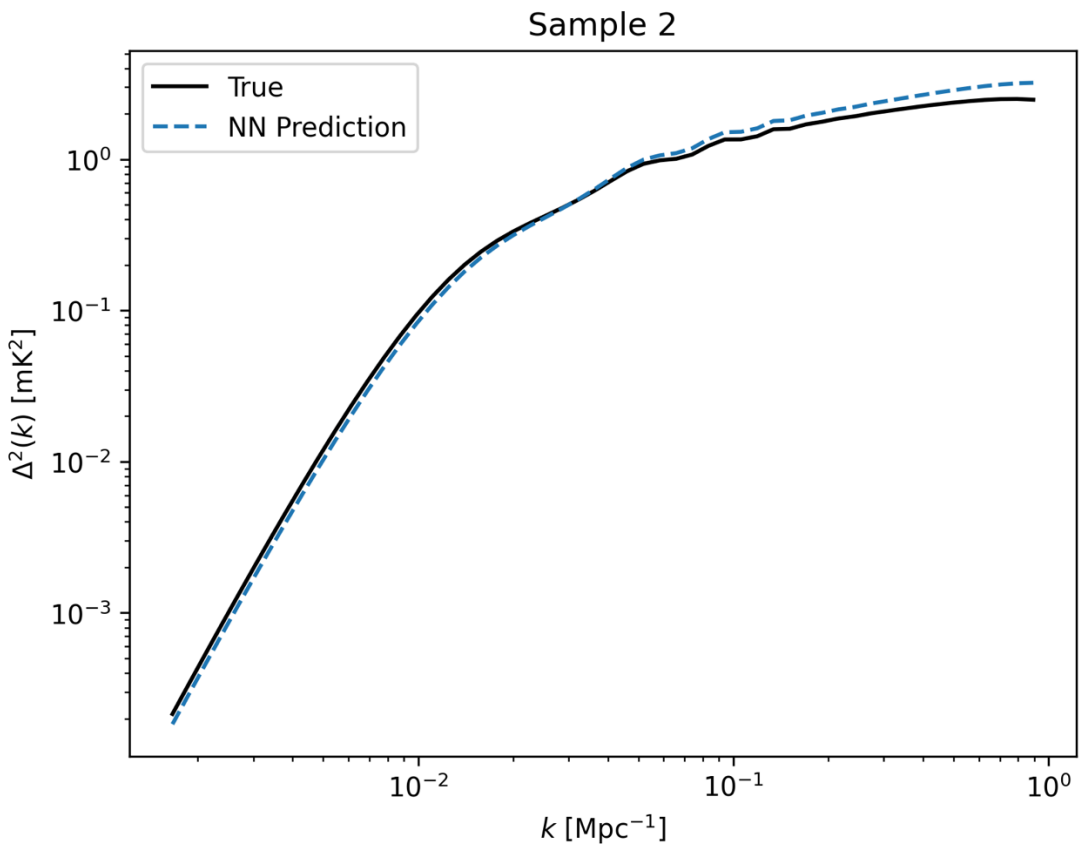
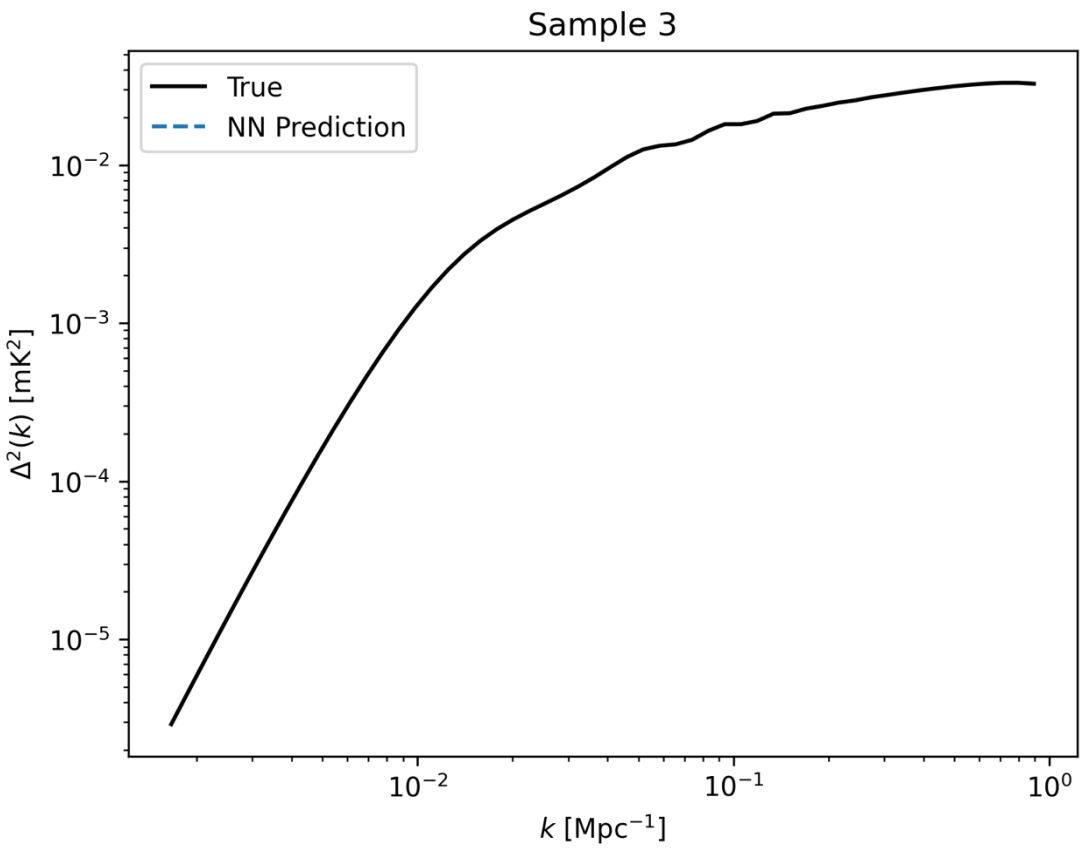
Assessing the accuracy





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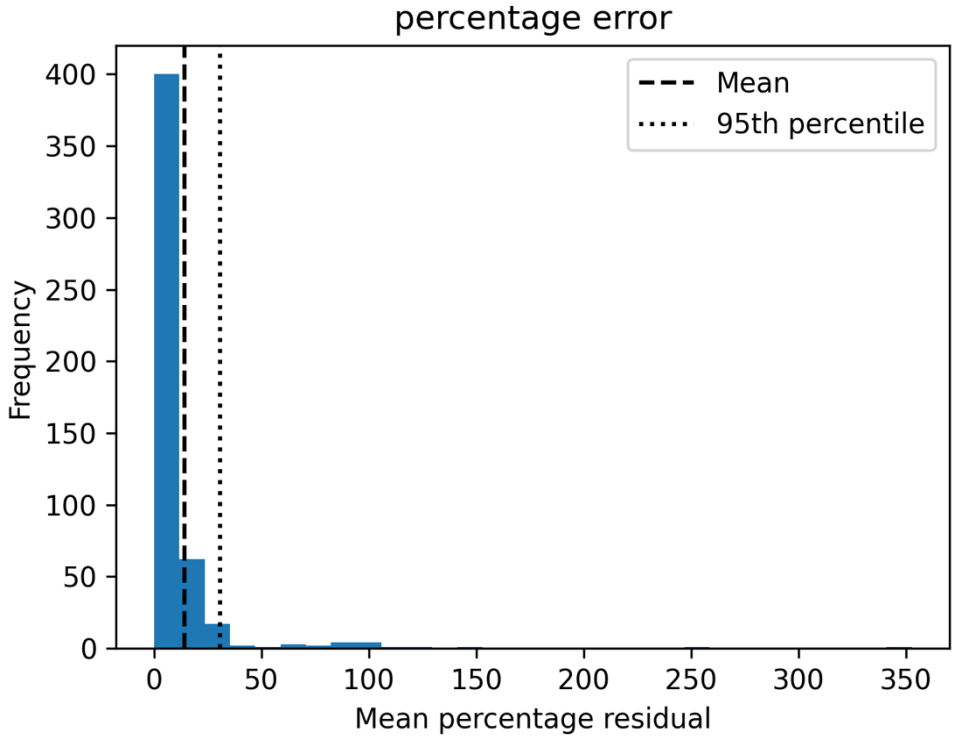
Comparing with the GP

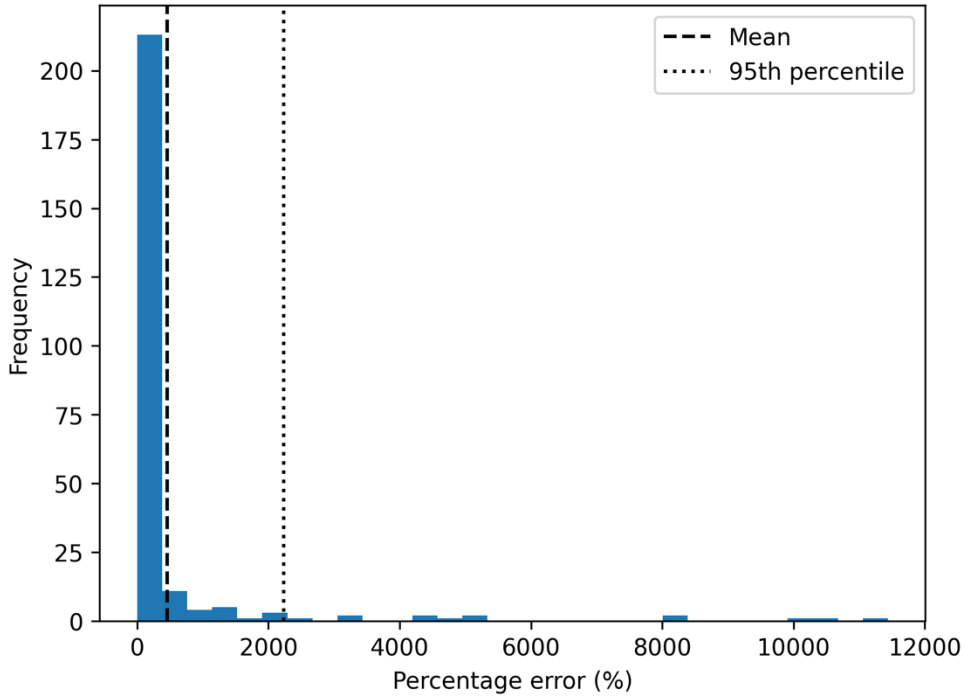
GP:

- Mean percentage error: 14.12 %
- 95th percentile: 30.82%



NN:

- Mean error: 456.693 %
- 95th percentile: 2230.327 %



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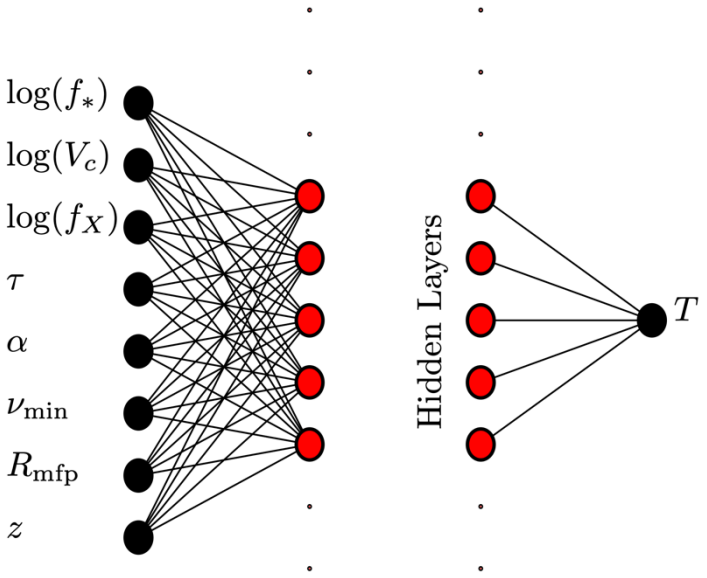
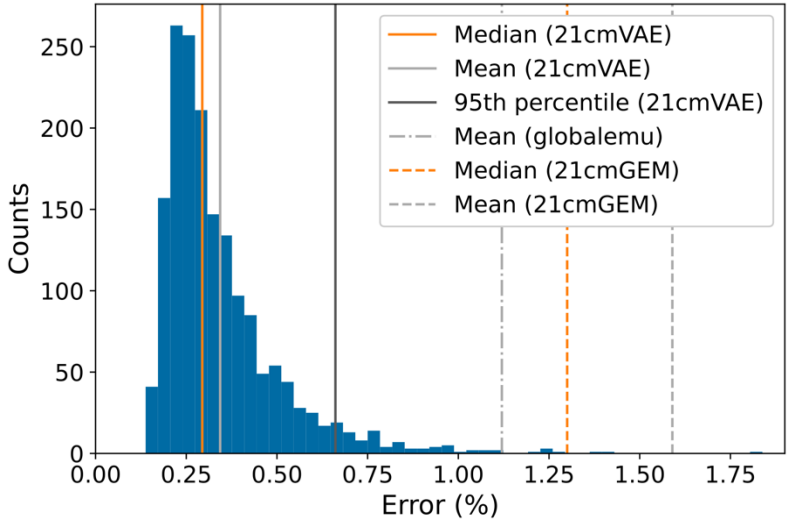
Current state of the art

- globalemu [2104.04336] uses small MLPs
- 21cmVAE [2107.05581] uses Variational Autoencoders
- 21cmLSTM [2410.07619] uses long and short term memory networks
- 21cmKAN [2508.11752] uses Kolmogrov-Arnold networks





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Summary

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Emulators allow us to do inference

- Emulators are an efficient way to approximate complex simulations
- They take of order milliseconds to evaluate compared to minutes, hours or even days per realization making inference possible
- Sensible choices about architectures, use of dimensionality reduction techniques, loss functions etc can make a big difference to the run time and accuracy of emulators

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Many different ways to do this

- Lots of different ways to build emulators
  - Here we have looked at Dense Neural Networks and Gaussian Processes
  - But we can also do this with long and short term memory networks, Kolmogrov-Arnold networks, variational autoencoders, convolutional neural networks…

Slides available at https://github.com/htjb/Talks (and Moodle!)
Example codes on github too!

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Next Lecture: Simulation Based Inference

- Why not just emulate the whole likelihood?